

# GUANZHANG LIU

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## SUMMARY

Data scientist with ~2 years of experience in EV battery intelligence at Li Auto, specialising in large-scale ML systems, feature engineering, and causal inference. Previously at ByteDance. Seeking data scientist roles in London.

## EXPERIENCE

### Li Auto

*Data Scientist, Battery Intelligence Department*

*Beijing, China*

Jul 2024 – Present

- Built a quasi-causal SOH degradation analytics system for Li Auto's million-scale fleet, identifying true lifespan drivers and quantifying that targeted usage adjustments extend battery life by ~30%; productised into an AI-powered self-service platform across multiple vehicle models, projecting tens of millions of RMB in warranty savings for a single vehicle model.
- Identified thermal runaway risk across a 1M+ vehicle fleet by reframing extreme class imbalance as a white-box risk-ranking problem; deployed a cross-platform ML scoring pipeline across 5 battery platforms with per-vehicle explainable outputs, enabling targeted after-sales inspection prioritisation.
- Productised battery behaviour feature engineering onto an internal ML platform — custom event definitions, multi-operator aggregations, and cross-platform normalisation across 5 battery platforms — improving iteration efficiency by ~85%.
- Developed a hybrid NLP pipeline for battery quality work-order classification — rule-based matching, TF-IDF, and Random Forest across 30+ fault categories; achieved ~90% accuracy on 15k+ samples, reducing manual labelling to confidence-guided hard-case review.
- Delivered battery analytics across BMS thermal strategy validation, user behaviour profiling, and life test optimisation (W02/X01) — building reusable end-to-end pipelines across 3+ vehicle platforms, directly shaping R&D strategy and product validation standards.

### ByteDance

*Data Analytics Intern, Douyin Market Strategy*

*Beijing, China*

May – Sep 2023

- Built a multi-source IP popularity scoring system integrating 5+ data platforms with ML-enhanced pipeline monitoring and workflow automation; achieved ~60% dashboard adoption and validated ~40% consumption growth following targeted campaigns.
- Delivered content strategy analytics for Qishui Music — redesigned music hotness metrics, built custom vertical classification frameworks, and developed multi-dimensional trending topic scoring to inform content selection and marketing resource allocation.

### Dongwu Securities

*Equity Research Intern, Mechanical Sector*

*China*

Dec 2021 – Mar 2022

- Produced an equity research report on Harmonic Drive Systems — multi-dimensional analysis across company fundamentals, competitive positioning, and market dynamics; built a stock price forecasting model and delivered short, medium, and long-term investment recommendations.

## EDUCATION

### Nanjing University

*MSc Computational Materials Science*

Sep 2021 – Jun 2024

- Developed Skeaf — a 3D Fermi surface visualisation tool integrating HDBSCAN clustering with quantum oscillation data; first author *Physical Review B* (2024); co-author *Physical Review B* (2024) & *Physical Review Materials* (2023); Patent: CN115795266A

### Nanjing University

*BSc Materials Science & Engineering (Outstanding Graduate) · GPA: 4.6/5.0 · Top 1% of cohort*

Sep 2017 – Jul 2021

- National Scholarship (top 0.15%) · MCM/ICM 2020 H Award (Team Captain) · Sakura Plan Exchange, Tohoku University, Japan (5 selected university-wide)

## SKILLS

|                            |                                                                                                                                                                      |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>         | Python · SQL/HiveSQL · C++ · MATLAB                                                                                                                                  |
| <b>ML &amp; Statistics</b> | Machine Learning · NLP · Feature Engineering · Causal Inference · Statistical Modelling · Time Series Analysis · LightGBM · scikit-learn · Transformer Models (BERT) |
| <b>AI Tools</b>            | AgentOps · Claude Code · OpenClaw (Self-hosted LLM) · LLM Integration · Prompt Engineering                                                                           |
| <b>Infrastructure</b>      | Linux · Hadoop · PyQt5 · Mayavi                                                                                                                                      |
| <b>Domain</b>              | Battery Intelligence · EV Analytics · Scientific Computing                                                                                                           |
| <b>Languages</b>           | English (TOEFL 104) · Chinese — Mandarin (Native) · Cantonese (Native) · Japanese (JLPT N1)                                                                          |